MLPerf(TM) Storage v3.0 - System Description

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Division: CLOSED
System name: rigA_c5n18xl
Workload: Training - 3D U-Net (unet3d), 1 simulated accelerator (b200), read_threads=24
Storage system: Alluxio Enterprise AI 3.9 (RC18) over AWS S3, via Alluxio-FUSE (POSIX)
Result: Accelerator Utilization 97.54% (5-run mean; bar >= 90%)
Cloud: AWS (us-west-1), single availability zone

1. Hardware configuration & node counts

Clients (host nodes): 1 x c5n.18xlarge
- 1 FUSE client, 1 simulated accelerator
- CPU: 72 vCPU (Intel Xeon Platinum 8124M/8259CL)   Memory: 192 GiB
- Storage: EBS gp3 (root)   Network: 100 Gbps ENA

Storage nodes: 1 x i3en.12xlarge
- 1 Alluxio storage worker (NVMe page store)
- CPU: 48 vCPU (Intel Xeon Platinum 8259CL)   Memory: 384 GiB
- Storage: 4x 7.5 TB NVMe SSD -> RAID0 (~28 TB) page store   Network: 50 Gbps ENA

2. System & network topology

Benchmark access is POSIX/file via Alluxio-FUSE; AWS S3 is the under-file-system (UFS). All nodes in one AWS us-west-1 availability zone over 50/100 Gbps ENA.

```
1x c5n.18xlarge  --Alluxio FUSE (POSIX, read)-->  1x i3en.12xlarge worker
                                                    NVMe RAID0 page store
                                                             |
                                                         AWS S3 (UFS)
```

3. Software stack & versions

- Alluxio Enterprise AI 3.9 (RC18) (distributed cache)
- Alluxio-FUSE (POSIX client)
- etcd 3.5.x (mount table + worker membership)
- Ubuntu 24.04 LTS
- OpenJDK 17
- mlpstorage 3.0.x (uv-managed) + DLIO 3.0.x
- Backing store: AWS S3 (UFS)

4. Non-default configuration options

- Client access: read-only (no client-side write cache / FDB)
- alluxio.dora.client.ufs.fallback.enabled=false; alluxio.client.list.status.from.ufs.enabled=false
- alluxio.user.fuse.enforce.direct.io.read=true; alluxio.user.fuse.random.access=true
- async prefetch: thread=512, part.length=4MB, max.part.number=8, file.length.threshold=4MB
- worker.network.netty.file.transfer=TRANSFER; frame.decoder.buffer.cumulator=COMPOSITE
- FUSE JVM: -Xms32g -Xmx32g -XX:MaxDirectMemorySize=32g
- Worker: write.cache.enabled=true + FoundationDB (worker-side; client is read-only)

5. Notes

Simulated accelerators (no GPU). Machine-readable system parameters are in the companion systemname.yaml. CLOSED workload parameters per mlpstorage; storage-system configuration is unconstrained in CLOSED. Dataset backing store: AWS S3.